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A meta-analysis of the effects of dietary protein concentration and degradability on milk protein yield and milk N efficiency in dairy cows

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ABSTRACT

Data sets from North American (NA, 739 diets) and North European (NE, 998 diets) feeding trials with dairy cows were evaluated to investigate the effects of dietary crude protein (CP) intake and ruminal degradability on milk protein yield (MPY) and efficiency of N utilization for milk protein synthesis (MNE; milk N ÷ N intake) in dairy cows. The NA diets were based on corn silage, alfalfa silage and hay, corn and barley grains, and soybean meal. The NE diets were based on grass silage, barley and oats grains, and soybean and rapeseed meals. Diets were evaluated for rumen-degradable and undegradable protein (RDP and RUP, respectively) concentrations according to NRC (2001). A mixed model regression analysis with random study effect was used to evaluate relationships between dietary CP concentration and degradability and MPY and MNE. In both data sets, CP intake alone predicted MPY reasonably well. Addition of CP degradability to the models slightly improved prediction. Models based on metabolizable protein (MP) intake predicted MPY better than the CP or the CP-CP degradability models. The best prediction models were based on total digestible nutrients (TDN) and CP intakes. Similar to the MPY models, inclusion of CP degradability in the CP (intake or concentration) models only slightly improved prediction of MNE in both data sets. Concentration of dietary CP was a better predictor of MNE than CP intake. Compared with the CP models, prediction of MNE was improved by inclusion of TDN intake or concentration. Milk yield alone was a poor predictor of MNE. The models developed from one data set were validated using the other data set. The MNE models based on TDN and CP intake performed well as indicated by small mean and slope bias. This meta-analysis demonstrated that CP concentration is the most important dietary factor influencing MNE. Ruminal CP degradability as predicted by NRC (2001) does not appear to be a significant factor in predicting MPY or MNE. Data also indicated that increasing milk yield will increase MNE provided that dietary CP concentration is not increased, but the effect is considerably smaller than the effect of reducing CP intake.

Key words: meta-analysis, milk protein yield, milk nitrogen efficiency

INTRODUCTION

Proper determination of animal protein requirements is critically important for maximizing production and minimizing N input in dairy production systems. Although, depending on the basal diet, increasing N input may produce an increase in milk protein yield (**MPY**), the efficiency of conversion of dietary N into milk protein will predictably decrease (Colmenero and Broderick, 2006). In cases where increased MPY was observed, this was usually a result of increased DMI (Broderick, 2003).

Previous meta-analyses indicated a strong relationship between diet composition variables and intake of specific nutrients and MPY in dairy cows (Hristov et al., 2004, 2005; Schwab et al., 2005; Huhtanen et al., 2008a). In these analyses, dietary CP and estimated MP (NRC, 2001), but not CP degradability, were significant predictors for MPY (Hristov et al., 2004). In a consequent analysis (Hristov et al., 2005), intakes of RDP or RUP were significant predictors for MPY in dairy cows. In a recent meta-analysis of a large data set (Huhtanen et al., 2008b), rumen protein balance and dietary CP concentration were the best predictors of milk N efficiency. In a comparison of several protein evaluation systems using a relatively small data set (n = 79), the German system (GfE, 1986), in which the supply of feed MP is estimated from non-urea CP using a fixed coefficient for CP, predicted MPY responses bet-

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Table 1. Dietary variables and animal parameters for North American (NA; n = 736) and North European (NE; n = 998) data sets used in the meta-analysis

	NA data				NE data			
Item	Mean	SD	Minimum	Maximum	Mean	SD	Minimum	Maximum
Diet composition, g/kg								
CP	178	22.1	103	208	165	21.5	101	252
RDP	122	21.2	57	221	120	16.5	78	189
Degradability[1]	0.68	0.063	0.41	0.83	0.73	0.044	0.54	0.83
RUP	56	11.8	28	106	45	9.7	20	86
RUP Digestibility	0.80	0.043	0.60	0.89	0.74	0.044	0.59	0.89
MP	103	10.8	69	150	94	9.1	64	132
NDF	306	48.2	182	537	407	50.0	195	571
NFC	416	56.3	174	562	305	52.4	80	520
TDN[2]	653	23.7	559	745	672	18.8	615	732
Nutrient intake/supply, kg/d								
DM	22.0	2.62	14.2	29.9	17.9	2.82	9.9	25.2
CP	3.9	0.68	2.0	6.4	3.0	0.59	1.3	5.1
RDP	2.7	0.55	1.0	4.3	2.15	0.41	1.03	3.83
RUP	1.2	0.29	0.5	2.3	0.81	0.23	0.25	1.85
MP	2.3	0.36	1.3	3.4	1.67	0.31	0.80	3.04
MPBact[3]	1.2	0.14	0.6	1.6	0.99	0.14	0.56	1.36
TDN	14.3	1.60	9.4	18.8	12.0	1.73	7.1	16.4
Animal parameters								
Milk yield, kg/d	31.4	5.41	20.1	46.4	25.4	5.05	13.0	45.8
Milk protein, g/kg	31.0	2.15	25.6	39.0	32.1	1.70	25.9	37.8
Yield, g/d	969	147.6	586	1,415	814	171.1	359	1,449
MNE,[4] g/kg	247	41.1	140	453	277	36.0	164	402
Milk fat, g/kg	34.2	3.84	21.1	46.1	42.8	3.96	31.9	55.0
Yield, g/d	1,068	189.9	572	1,662	1,081	217.0	479	1,671
BW,[5] kg	602	46.2	476	723	564	41	440	673
DIM, d	87	44.5	5	247	103	30.2	38	226

[1]Ruminal degradability of dietary CP (RDP ÷ CP).

[2]Total digestible nutrients.

[3]MP from bacterial protein.

[4]Milk N efficiency (milk N ÷ N intake).

[5]NA data set, n = 532 for BW; NE data set, n = 998 for BW.

ter than most of the other systems using feed-specific degradability and digestibility of RUP (Schwab et al., 2005). Earlier, Tuori et al. (1998) reported that using constant degradability coefficients predicted MPY responses within a study better than in situ-determined degradability estimates. Ipharraguerre and Clark (2005) demonstrated little effect of increasing RUP intake on production in dairy cows.

Based on these analyses, it appears that the role of dietary CP degradability as a determinant of MPY is overvalued in the current feeding models for dairy cows. Furthermore, increasing public pressure for sustainable animal production necessitates reexamination of the factors that are essential in optimizing the efficiency of transfer of dietary N into milk protein (milk N efficiency, **MNE**) in dairy cows. The objective of the present meta-analysis was to evaluate the effects of dietary CP intake and ruminal degradability predicted according to NRC (2001) on MPY and MNE using 2 large data sets based on North American and North European feeding trials.

MATERIALS AND METHODS

Experimental Data

North American Data Set. The North American (**NA**) data set (Hristov et al., 2004, 2005) represented 736 treatment means (diets) from 218 feeding trials conducted in the United States and Canada involving Holstein cows and published in the *Journal of Dairy Science* (volumes 73 through 83). Body weights of the cows were available for 532 treatment means. The diets were based on corn silage, alfalfa silage and hay, corn and barley grains, and soybean meal. Nutrient composition of the diets was estimated using the NRC (2001) model. Nutrient intakes were estimated based on published DMI and NRC-derived compositions. All diets were fed as TMR and represented a wide range of CP and RDP or RUP concentrations; the intake of these nutrients varied markedly among diets (Table 1).

North European Data Set. The North European (**NE**) data analyzed in this study were collected from 207 lactation trials including 998 treatment means. The list of references used to collect the data is provided in a data supplement available online (http://jds.fass.org/content/vol92/issue7/); the studies were conducted from 1979 to 2005. Grass silage was the main forage source, but for 82 and 38 diets, respectively, grass silage was partially or completely replaced with legume (mainly red clover) and whole-crop silages (barley, wheat, corn). Forages were fed ad libitum in all trials, and concentrates were fed on a flat-rate basis irrespective of milk yield (**MY**). The main ingredients in concentrates were barley, oat, by-products from the food industry, canola meal, and soybean meal. The prerequisite for an experiment to be included in the data set was that silage and total DM intake, milk production, and adequate forage characterization (plant species, DM concentration, in vivo or in vitro digestibility, and fermentation quality) were reported.

Estimation of NRC Feeding Parameters

Composition of the NA diets was estimated using NRC (2001) software. Feeds listed in the publications were matched with feeds from the feeding model libraries. The chemical composition of concentrate feeds was not edited. Where available, CP and NDF concentrations of forages were modified to reflect actual, published, dietary concentrations. Details on the composition of the diets are given in Hristov et al. (2004, 2005).

For the NE data set, concentration of total digestible organic matter (**DOM**; g/kg of DM) at maintenance level of feeding was estimated as described by Huhtanen et al. (2008b). Briefly, DOM concentration for concentrates was estimated using tabulated digestibility coefficients and analyzed chemical composition. For silages, DOM concentration was estimated either in vivo in sheep fed at maintenance level (n = 467), in vitro using rumen fluid (n = 114), a pepsin-cellulase method (n = 224), or by some other laboratory method (n = 183). The same method was used within a study.

Total digestible nutrients (**TDN**) at maintenance level of feeding (**TDN_m**; g/kg of DM) were calculated from DOM as follows: TDN_m = DOM + 1.25 × fatty acids (g/kg of DM) to correct for the greater energy concentration of fat compared with the other chemical components. Concentrations of TDM at production level (**TDN_p**) were calculated according to NRC (2001) using the following equation to discount for the reduced digestibility with increased feeding level: $TDN_p = TDN_m - [0.18 \times TDN_m - 103) \times (FL - 1)]$, where FL = feeding level, multiple of maintenance. Feeding level was calculated using the requirements of 0.123 Mcal of ME/kg of $BW^{0.75}$ and 1.23 Mcal/kg of ECM for maintenance and milk production, respectively (MTT, 2006).

Concentrations of RPD, RUP, and digestible RUP were calculated using the NRC (2001) tabulated values for CP degradation parameters and digestibility of RUP. For grass silages, the parameter values tabulated by NRC (2001) were selected based on the nearest NDF concentration. In the NE data set, legume silages were mostly red clover, which is less degradable than alfalfa or grass silages (Dewhurst et al., 2003; Brito et al., 2007); therefore, degradation parameters used for red clover were adjusted to give about 10-percentage-units lower degradability compared with grass silages.

For both data sets, effective protein degradability was calculated using passage rates estimated from the data using NRC (2001) equations.

Yield of microbial CP (**MCP**; g/d) was calculated as 130 × TDN_p (kg/d) when RDP intake exceeded 1.18 × MCP yield (NRC, 2001). When RDP intake was less than 1.18 × TDN-predicted MCP yield, MCP was calculated as 0.85 × RDP intake; that is, in such diets, the amount of MP from bacterial protein (**MPBact**) was determined by RDP supply; MPBact was calculated as 0.64 × MCP. Total MP and MP derived from ruminally undegraded feed protein (**MPFeed**) were calculated as described by NRC (2001). Milk N efficiency was defined as milk N ÷ N intake.

Statistical Methods

The relationships between N utilization and animal or dietary measurements within an experiment were investigated using the MIXED procedure of SAS (Littell et al., 1996) and the following model:

$$Y = B_0 + B_1X_{1ij} + b_0 + b_1X_{1ij} + B_2X_{2ij} + B_3X_{3ij} + e_{ij}, \quad [1]$$

where $B_0 + B_1X_{1ij} + B_2X_{2ij} + B_3X_{3ij}$ are the fixed effects and b_0, b_1, and e_{ij} are the random experiment effects (intercept, slope, and error), i = 1...218 (NA data set) or 1...207 (NE data set) studies, and j = 1...n_i values.

Because models did not converge when the slope was random and several independent variables were used, it was decided to use only random intercept and random slope statement for the first independent variable for all models. The models were first constructed based on a single parameter and then other biologically relevant variables were included. Any additional parameter was assumed to be a fixed factor. The goodness of fit of the models was evaluated using residual mean square error (**RMSE**), which was calculated as a square root of residual variance. The ratio between standard error of random slope (square root of error variance) divided by slope was computed to indicate variability of the random slope. Akaike's information criterion (**AIC**) was used to evaluate the variance-covariance structure. Adjusted values were calculated for the figures by adding the residual from each individual observation to the predicted value of the study regression (St-Pierre, 2001).

Linear and multiple regression models were used to develop prediction equations for MPY and MNE from both data sets. An evaluation was undertaken to validate the prediction equations derived from NE data using the NA data, and likewise, the equations derived from the NA data were validated using the NE data. Prediction accuracy of relationships was examined using the mean square prediction error (**MSPE**): MSPE $= \sqrt{[1/n \times \Sigma \text{ (observed} - \text{predicted)}^2]}$, where n is the number of observations. Mean squared prediction error was divided into components resulting from mean bias, slope bias, and random variation around the regression line (Bibby and Toutenburg, 1977).

RESULTS

Chemical compositions of the diets used in the analysis, nutrient intakes, and animal parameters are shown in Table 1. Corn silage, alfalfa silage, and hay were the predominant forage components of the NA diets, whereas grass silage was the major forage component of the NE diets. Corn and barley grains were the main energy concentrates of the NA diets. Cereal grains, mainly barley and oats, comprised on average 76% of the energy concentrates of the NE diets. Solvent-extracted soybean meal (NA and NE) and rapeseed meal (NE) were the main protein supplements.

Both data sets showed a large variation in chemical composition of diets. The mean CP concentration was greater in the NA diets compared with the NE diets (178 vs. 165 g/kg of DM), the difference being almost exclusively in RUP content. The variability in RUP concentration was larger in the NA diets. The mean concentration of NDF was greater in the NE diets than in the NA diets (407 vs. 306 g/kg of DM, respectively), whereas an opposite difference was observed for NFC.

The concentration of MP as specified by NRC (2001) was greater in the NA than in the NE diets (103 vs. 94 g/kg of DM). The difference was mainly because of the greater RUP concentration and partly because of the greater RUP digestibility.

Diets represented a wide range in DMI, which was 4 kg/d greater with the NA diets than with the NE diets. Differences in the intake of protein fractions reflected differences in DMI and concentration of protein fractions. The NA cows produced 6 kg/d more milk than the NE cows, but protein and especially fat concentrations were lower in the NA cows; ECM yields were 28.5 versus 26.1 kg for the NA and NE cows, respectively, and milk efficiencies were 1.30 versus 1.45 kg of ECM/kg of DMI. Milk N efficiency was 12% greater in the NE than in the NA cows (277 vs. 247 g/kg). Cows in the NA data set were, on average, about 38 kg heavier than cows in the NE data set.

Prediction of MPY

NA Data Set. Variables evaluated in this analysis were chosen based on our previous analyses (Hristov et al., 2004, 2005; Huhtanen et al., 2008b) and the objective of this study. Crude protein intake alone predicted MPY reasonably well in the NA data set (Table 2). Addition of CP degradability to the model slightly improved prediction (as judged by the RMSE). Models based on MP intake (linear or quadratic) predicted MPY better than the CP or the CP-CP degradability models. Compared with MP, MPBact was a slightly better predictor of MPY in the NA data set although the coefficient of variation (**CV**) of random slopes was similar between the 2 models. Prediction error of the MPBact model was reduced by including MPFeed or CP in the model. The slope of MPBact was about 5-fold greater compared with MPFeed. Models based on TDN intake predicted MPY better than models based only on protein variables.

NE Data Set. Residual mean square errors of all models were smaller in the NE data than in the NA data, but the ranking of the models based on RMSE was consistent between data sets. Using degradability as a variable with CP intake or separating CP intake into RDP and RUP intakes caused only small improvements in the model, according to RMSE. Similar to the NA data, MP intake predicted MPY responses better than CP intake. The quadratic model of MP intake was better than the linear model and showed diminishing production responses with intake. Bacterial MP and TDN predicted MPY as well as total MP, but the CV of random slope was smaller for the MPBact and TDN models. Similar to the NA data set, prediction error of the MPBact model was markedly reduced by including MPFeed or CP in the model. As with the NA data, the slope of MPBact was greater than that with MPFeed. Variables included in the best model were TDN and CP intake and degradability.

Prediction of MNE

NA Data Set. Similar to the MPY models, inclusion of CP degradability in the CP (intake or concentration) models had negligible effect on MNE prediction (based on RMSE) in the NA data set (Table 3).

Table 2. Analysis of dietary intake variables important in predicting milk protein yield (g/d) in dairy cows

X_1	X_2	X_3	Intercept	SE	$Estimate_1$	SE	$Estimate_2$	SE	$Estimate_3$	SE	RMSE[1]	AIC[2]	CV[3]
NA data[4,5]													
CP			560	33.0	103	8.3					48.8	8,632	0.579
CP	Degr[6]		736	52.4	103	8.3	−252	58			47.9	8,604	0.594
RDP	RUP		596	31.4	75.6	10.7	138	12			48.6	8,610	1.003
MP			491	30.9	209	13.5					40.5	8,471	0.517
MP	MP × MP		114	110.0	546	95.5	−74.0	20.8			39.8	8,450	0.201
MPBact[7]			266	44.9	589	37.6					39.6	8,414	0.535
MPBact	MPFeed[8]		289	43.2	491	38.1	95.7	12.6			37.4	8,352	0.610
MPBact	CP		274	45.5	501	45.8	24.6	7.2			39.1	8,397	0.644
TDN			162	47.5	55.9	3.30					38.1	8,371	0.471
TDN	CP		175	48.2	47.7	3.81	27.0	6.08			37.4	8,347	0.576
TDN	CP	Degr	277	58.9	47.6	3.78	26.1	6.11	−136	46.6	37.1	8,328	0.564
NE data[9]													
CP			354	19.3	153	6.3					25.8	10,570	0.360
CP	Degr		507	42.3	149	6.3	−193	47.7			25.6	10,544	0.369
RDP	RUP		368	18.7	137	9.3	176	11.6			26.2	10,564	0.508
MP			256	18.8	329	11.5					23.1	10,327	0.315
MP	MP × MP		−129	46.4	793	53.9	−139	15.9			22.6	10,262	0.141
MPBact			−89	22.7	897	24.2					23.8	10,188	0.222
MPBact	MPFeed		−28	22.4	750	25.8	142	10.8			21.4	10,024	0.262
MPBact	CP		−24	23.4	646	29.3	62.5	4.23			20.5	9,995	0.321
TDN			−100	23.5	75.5	2.08					23.7	10,197	0.232
TDN	CP		−31	23.5	54.0	2.44	63.5	4.14			20.4	9,988	0.323
TDN	CP	Degr	105	35.6	54.0	2.41	60.4	4.14	−175	34.5	20.2	9,954	0.287

[1]Root mean square error (square root of residual variance).

[2]Akaike's information criterion.

[3]Coefficient of variation of random slope (square root of slope variance/slope).

[4]North American data set (n = 736).

[5]CP, RDP, RUP, MP, DM, MPBact, and TDN (total digestible nutrients) in kg/d.

[6]Ruminal degradability of dietary CP (RDP/CP); g/g.

[7]MP from bacterial protein.

[8]MP from feed protein.

[9]North European data set (n = 998).

Coefficients for CP (and RDP-RUP) intakes or dietary concentrations were negative, reflecting the negative effect of increasing dietary CP intake or concentration on MNE in dairy cows. Milk yield was a significant ($P < 0.001$) but poorer predictor of MNE (RMSE = 19.5 g/kg and AIC = 7,079; model not shown) compared with CP intake (RMSE = 12.8 and AIC = 6,666), or dietary CP concentration (RMSE = 12.5 g/kg and AIC = 6,594). Inclusion of MY in the CP concentration model, however, significantly improved the precision of MNE prediction (a reduction in RMSE of 2.8 g/kg) and reduced the CV of the random slope. The coefficient for MY was positive, reflecting the increase in MNE with increasing MY. Inclusion of CP degradability to these models further slightly improved prediction (intake and concentration models, respectively; models not shown), but rendered CP degradability insignificant ($P = 0.22$). Compared with the CP-CP degradability models, prediction of MNE was significantly improved by inclusion of TDN intake or concentration. No further improvement of MNE prediction was achieved by inclusion of CP degradability in the TDN-CP models. Inclusion of NDF (but not NFC; $P = 0.40$) in the TDN-CP intakes-CP degradability model improved (F-test of model comparison, $P < 0.001$) MNE prediction. The coefficient for NDF intake was negative (−4.86 kg/d; model not shown). The effects of NDF or NFC concentrations on MNE were not significant when included separately in the TDN-CP concentration-CP degradability model ($P = 0.22$ and 0.93, respectively).

NE Data Set. Crude protein concentration predicted MNE better than CP intake in the NE data set. The effect of CP degradability on MNE was significant when included as a second variable with CP intake or CP concentration, but RMSE was reduced only marginally. Similar to the NA data, MY was not related ($P = 0.30$) to MNE when used as a single factor in a model. However, MY was positively associated with MNE when included in the CP concentration model. The best models, based on either intake or concentration data, included TDN, CP, and CP degradability as independent variables. The effects of starch and NFC concentrations on MNE were positive ($P < 0.001$) and that of NDF negative ($P < 0.01$), when added sepa-

Table 3. Analysis of dietary variables important in predicting milk N efficiency (g/kg) in dairy cows

X_1	X_2	X_3	Intercept	SE	$Estimate_1$	SE	$Estimate_2$	SE	$Estimate_3$	SE	RMSE[1]	AIC[2]	CV[3]
NA data[4]													
Intakes[5]													
CP			393	10.6	−36.9	2.6					12.8	6,666	0.607
CP	Degr[6]		435	15.1	−37.0	2.5	−60.8	15.6			12.6	6,643	0.602
RDP	RUP		401	9.3	−43.5	3.1	−28.0	3.3			13.2	6,658	0.573
CP	TDN		280	11.3	−61.4	2.6	14.5	0.83			10.3	6,415	0.328
CP	TDN	Degr	304	15.0	−61.1	2.6	14.2	0.84	−31.0	13.1	10.2	6,402	0.325
Concentrations[7]													
CP			496	14.0	−1.40	0.1					12.5	6,594	0.371
CP	Milk yield		402	12.8	−1.54	0.1	3.72	0.201			10.1	6,314	0.287
CP	Degr		529	17.2	−1.40	0.1	−47.8	14.8			12.4	6,576	0.362
RDP	RUP		487	11.5	−1.43	0.1	−1.14	0.075			12.5	6,583	0.375
CP	TDN		183	30.4	−1.28	0.1	0.45	0.039			11.4	6,478	0.401
CP	TDN	Degr	194	34.8	−1.29	0.1	0.44	0.040	−8.9[8]	14.1	11.4	6,470	0.398
NE data[9]													
Intakes													
CP			392	7.2	−39.8	2.2					8.64	8,401	0.498
CP	Degr		436	14.6	−41.0	2.2	−56.8	16.1			8.56	8,381	0.494
RDP	RUP		397	7.1	−45.9	3.2	−31.0	3.95			8.76	8,403	0.560
CP	TDN		258	7.8	−71.6	2.3	18.9	0.72			7.07	7,924	0.272
CP	TDN	Degr	290	12.9	−72.1	2.3	18.8	0.72	−38.9	12.7	6.67	7,875	0.268
Concentrations													
CP			475	7.4	−1.21	0.040					7.91	8,036	0.267
CP	Milk yield		421	7.8	−1.36	0.042	3.15	0.18			7.26	7,806	0.264
CP	Degr		563	22.3	−1.26	0.042	−101	23.9			7.85	8,010	0.263
RDP	RUP		479	7.6	−1.28	0.059	−1.14	0.063			7.75	8,035	0.370
CP	TDN		167	28.3	−1.22	0.038	0.46	0.041			7.22	7,925	0.267
CP	TDN	Degr	166	27.8	−1.24	0.038	0.54	0.043	−72.5	13.7	7.15	7,891	0.256

[1]Root mean square error (square root of residual variance).
[2]Akaike's information criterion.
[3]Coefficient of variation of random slope (square root of slope variance/slope)
[4]North American data set (n = 736).
[5]CP, RDP, RUP, and TDN (total digestible nutrients) in kg/d.
[6]Ruminal degradability of dietary CP (RDP/CP), g/g.
[7]CP, RDP, RUP, TDN in g/kg; Degr in g/g.
[8]The effect of Degr is not significant ($P > 0.10$).
[9]North European data set (n = 998).

rately in the TDN, CP concentration, and CP degradability models (models not shown). However, although significant, the quantitative effects of these parameters on MNE were small (0.051, 0.035, and −0.028 g/kg DM for starch, NFC, and NDF, respectively).

Validation of the Models

The validations of the prediction models developed from the NA data set using the NE data set are shown in Table 4 and Figure 1. Mean prediction error for MPY was smallest for the quadratic model of MP intake. Mean bias was negative for all models. Including degradability in the TDN-CP model did not improve prediction. The slopes of all models were significantly greater than 1.00; that is, the NE cows responded to changes in nutrient supply better than predicted by the models derived from the NA data set. Except for the quadratic MP model, the errors resulted mainly from the mean and slope biases. The NA MNE models performed better in predicting MNE in the NE data set than the NA MPY models. The mean prediction errors of the models including concentrations or intakes of TDN and CP either without or with CP degradability were only slightly greater compared with the models derived from the NE data. Both the mean and slope biases were small and random error contributed most of the variation. The actual MNE was greater than predicted by models based on CP concentration or CP concentration and CP degradability as predictors.

Similar to the validation of NA models, the quadratic MP intake MPY model developed based on the NE data set and validated against NA diets had the lowest MSE with the random bias accounting for most of the prediction error (Table 5; Figure 1). The NE MNE models validated with the NA data set performed better than the MPY models. The least MSE and mean bias were associated with the TDN-CP concentration or intake (with or without CP degradability) models because most of the error was due to random bias.

DISCUSSION

The NA diets used in this analysis had on average about 8% greater CP concentration than the NE diets. This difference is mostly determined by the type of forages fed—alfalfa silage or hay (NA diets) versus grass silage (NE diets)—and to a smaller extent by differences in CP content of the protein supplements (soybean vs. rapeseed meals). These differences were partially compensated for by differences in CP concentration among the energy concentrates most commonly fed in the 2 data sets (corn vs. barley grain, for example). The most remarkable difference between the NA and NE diets was in the NDF and NFC concentrations. The greater NDF concentration in the NE diets reflects differences in NDF concentration between dietary ingredients rather than different forage-to-concentrate ratio. For example, NDF concentration is greater in barley and oats compared with corn (220–240 vs. 100 g/kg of DM), in rapeseed meal compared with soybean meal (270 vs. 120 g/kg), and in grass silages compared with alfalfa or corn silages. The mean proportion of forage NDF of the total NDF was similar in the NA and NE data sets (73 and 76%, respectively).

North American cows were heavier and consumed more dietary DM compared with NE cows, which reflects breed differences between the 2 data sets (Holstein vs. predominantly Ayrshire and Friesian in NA and NE, respectively). This greater DMI was reflected in greater milk production in the NA data set. Cows

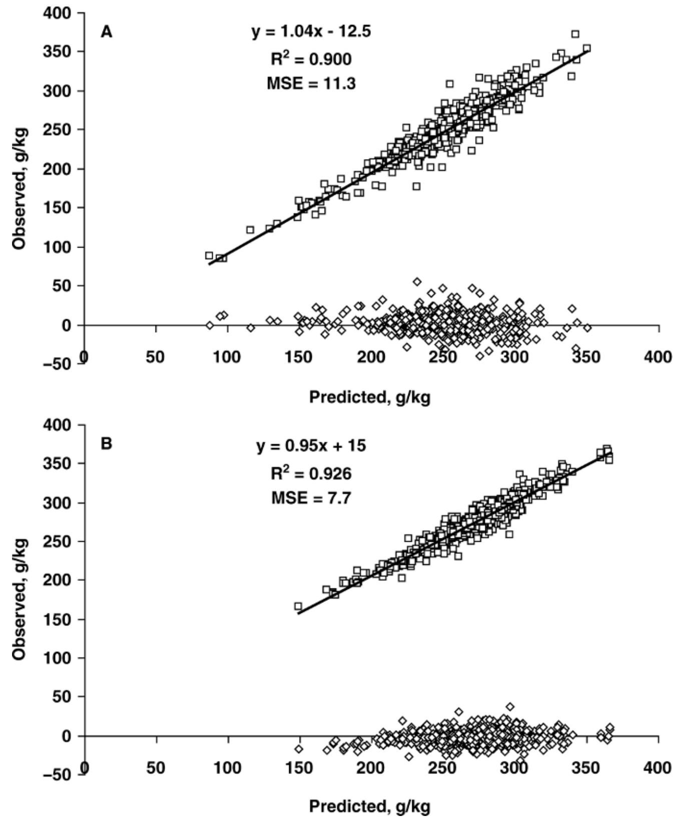

Figure 1. Validation of the milk N efficiency model [milk N efficiency (MNE) = total digestible nutrients (TDN) intake + CP intake) estimated from the North European (NE) data against the North American (NA) data (A) and the model estimated from the NA data against the NE data (B). The values are adjusted for the random study effect. Residuals (observed – predicted) are shown (◊).

Table 4. Validation of North American (NA) models using the North European (NE) data set

									Distribution of error variance		
X_1	X_2	X_3	Intercept	SE	Estimate	SE	MPE[1]	Mean bias	Bias	Slope	Random
MPY[2]											
MP[3]			−517	45.1	1.57	0.055	56.0	−34.6	0.381	0.450	0.169
MP	MP × MP		−110	28.5	1.12	0.037	28.9	−14.2	0.240	0.138	0.622
TDN[3]	CP[3]		−320	28.6	1.36	0.036	45.5	−21.2	0.216	0.573	0.211
TDN[3]	CP[3]	Degr[3]	−310	28.0	1.36	0.036	46.1	−21.1	0.210	0.561	0.229
MNE[4]											
CP[5]			46	7.1	0.86	0.028	13.2	9.8	0.546	0.096	0.358
CP[5]	Degr		46	7.2	0.87	0.029	14.6	11.6	0.636	0.069	0.295
TDN[5]	CP[5]		15	7.4	0.95	0.029	7.7	2.2	0.079	0.028	0.893
TDN[5]	CP[5]	Degr	15	7.4	0.96	0.029	7.8	2.6	0.114	0.024	0.862
TDN[3]	CP[3]		−32	9.9	1.12	0.036	8.1	1.7	0.044	0.119	0.837
TDN[3]	CP[3]	Degr	−36	9.9	1.14	0.037	8.4	2.6	0.097	0.139	0.765

[1]Mean prediction error.

[2]Milk protein yield in g/d.

[3]MP, TDN (total digestible nutrients), and CP intakes in kg/d; Degradability of dietary CP (Degr) in g/g.

[4]Milk N efficiency in g/kg.

[5]CP and TDN dietary concentrations in g/kg.

Table 5. Validation of North European (NE) models using North American (NA) data set

									Distribution of error variance		
X_1	X_2	X_3	Intercept	SE	Estimate	SE	MPE[1]	Mean bias	Bias	Slope	Random
MPY[2]											
MP[3]			329	41.2	0.64	0.041	69.8	−36.8	0.277	0.388	0.335
MP	MP × MP		−168	76.4	1.21	0.082	50.9	27.2	0.284	0.063	0.654
TDN[3]	CP[3]		219	43.6	0.75	0.044	55.9	−27.9	0.248	0.278	0.474
TDN[3]	CP[3]	Degr[3]	207	43.6	0.76	0.043	57.1	−32.4	0.321	0.247	0.432
MNE[4]											
CP[5]			−53	16.3	1.16	0.063	18.1	−12.2	0.451	0.029	0.520
CP[5]	Degr		−36	14.5	1.05	0.054	25.9	−22.6	0.752	0.003	0.245
TDN[5]	CP[5]		−13	12.5	1.04	0.051	11.3	−2.5	0.050	0.012	0.937
TDN[5]	CP[5]	Degr	11	11.4	0.94	0.046	12.0	−3.4	0.082	0.028	0.891
TDN[3]	CP[3]		32	10.7	0.87	0.044	12.1	−1.5	0.016	0.166	0.818
TDN[3]	CP[3]	Degr	34	10.4	0.85	0.043	12.4	−2.2	0.030	0.191	0.779

[1]Mean prediction error.

[2]Milk protein yield in g/d.

[3]MP, TDN (total digestible nutrients), and CP intakes in kg/d; Degradability of dietary CP (Degr) in g/g.

[4]Milk N efficiency in g/kg.

[5]CP and TDN dietary concentrations in g/kg.

in the NE data set, however, had greater component concentrations, which, combined with the 0.9 kg/d lower CP intake, resulted in about 12% greater MNE compared with cows in the NA data set.

In both data sets, TDN intake predicted MPY at least as well as MPBact, suggesting that NRC (2001) overestimates RDP requirements. Bacterial MP is a linear function of TDN intake when RDP requirements are met; it should predict MPY better than TDN if discounts for deficient RDP supply are correct. In agreement with the present study, Schwab et al. (2005) reported better predictions of MPY for the NRC (2001) and INRA (1989) systems without discounting MPBact for deficient RDP supply. NRC (2001) sets the RDP requirement to microbial N ratio to 1.18; that is, the efficiency of the capture of RDP by rumen microbes is assumed to be 0.85 with no provision for ruminal recycling of N. The review by Lapierre and Lobley (2001) indicated that substantial amounts of urea can be recycled to the rumen, and safety margins may not be needed to meet the RDP requirements of rumen microbes. Recently, Gozho et al. (2008) reported that 58 to 65% (depending on treatment) of the urea N synthesized in the liver was returned to the gastrointestinal tract of lactating dairy cows.

Better predictions of MPY using TDN intake as an independent variable in the present models indicate that feed MP was not accurately estimated by NRC (2001). The significant ($P < 0.001$) effect of CP degradability on MPY suggests that the tabulated degradation parameters reflect qualitative rather than quantitative differences between feeds. In agreement with results of the present study, which is based on more than 1,700 treatment means, Schwab et al. (2005) reported that the German feed protein evaluation system (GfE, 1986) predicted MPY responses better than NRC (2001) in a North American data set (n = 59) and in a Finnish (n = 72) data set. In the German model, the contribution of microbial MP is calculated from ME intake and feed MP using a constant coefficient for urea-free CP intake; that is, the model assumes constant degradability and digestibility of RUP. In the present study, the models with TDN and CP intake as independent factors predicted MPY better in both data sets than MP. These models use the same parameters as the German model and ignore the variation between feeds both in ruminal CP degradability and intestinal digestibility of RUP. Tuori et al. (1998) reported, based on 157 treatment means, that using a constant protein degradability (0.80) for all feeds predicted differences in protein value of the diet within a study better than using degradability values determined by the in situ method.

One possible reason for the minimal effects of CP degradability on MPY could be that MP supply exceeded requirements and the cows were not very responsive to increased MP supply resulting from reduced ruminal protein degradability. The coefficient for degradability in the 3-factor model was −136 or −175 (NA and NE data sets, respectively); that is, a 1-percentage-unit decrease in protein degradability would increase milk protein yield by 1.4 or 1.8 g/d (respectively). Assuming DMI of 22 kg/d of a diet containing 160 g of CP/kg of DM and 0.80 digestibility of RUP, a 1-percentage-unit decrease in degradability would correspond to 28 g/d increase in MP supply. Marginal utilization of incremental MP would then be 6.3%. When the data were restricted to diets in which NE_L-allowable MY was greater than MP-allowable MY to ensure that MP was more limiting, the corresponding marginal MP efficiency increased to 10.1%. This value is substantially lower than the values derived from postruminal casein infusion studies (Clark et al., 1977; Choung and Chamberlain, 1993). Consistent with the small responses to dietary CP degradability in both data sets, the slope of MPFeed was much lower (81%) than the slope of MPBact in the MPBact-MPFeed model. Overestimation of the range in degradability and feed MP supply are the most likely explanations for the poor production responses to increased feed MP supply. Greater variability of the random slope for the MP model compared with the TDN or MPBact models in the NE data set also suggest inaccuracies in estimation the MPFeed component.

In NRC (2001), RDP and RUP are calculated from the degradation kinetic parameters estimated by the in situ method using the following models: RDP = A + B $[K_d/(K_d + K_p)]$ and RUP = B $[K_p/(K_d + K_p)]$ + C, where all of fraction A is considered to be degraded, fraction B is degraded at a rate K_d, and fraction C is considered undegradable. The proportion of fraction B degraded is a function of K_d and the passage rate K_p. According to recent knowledge, these models appear to have 2 false assumptions: fraction A is not completely degraded in the rumen and passage of feed particles from the rumen cannot be described as a single first-order compartment.

In dairy cows fed alfalfa hay or silage diets, the soluble, nonprotein, nonammonia, nonmicrobial N represented 30 to 35% of the total N pool in the rumen and had a very high (7.7 g/h) outflow rate (Hristov et al., 2001). Omasal flow measurements, in vitro studies, and studies using ^{15}N-labeled feeds have also shown a considerable flow of feed-soluble nonammonia N (**SNAN**) with peptides being quantitatively the most important component (Choi et al., 2002; Hedqvist and Udén, 2006; Ahvenjärvi et al., 2007; Reynal et al., 2007). Consistent with the above-cited studies, the proportion of SNAN had no effect on MPY when included in a model with MP estimated using constant ruminal degradability and intestinal digestibility of RUP (Huhtanen et al., 2008a). This suggests that the proportion of SNAN (a large proportion of fraction A in situ) was not related to ruminal protein degradability and MP, as models estimating RUP from degradation kinetics suggest.

Marker kinetics data estimated from duodenal samples strongly indicate that the passage of feed particles cannot be described by assuming the rumen is a single first-order kinetic system (Ellis et al., 1994; Huhtanen et al., 2006). Both intrinsically (ADF-^{15}N; Huhtanen and Hristov, 2001) and extrinsically (Lund et al., 2006) labeled forages showed an ascending phase of marker excretion curves, and using the passage rate estimated from the descending phase of marker curves clearly underestimates the residence time in the rumen during which the feed is subjected to degradation. In the present meta-analysis, the mean passage rate of forages in the NE data averaged 0.05 per h (SD = 0.002) when calculated using the NRC (2001) equations. The corresponding rumen residence time (20 h) is markedly shorter than estimated from duodenal marker excretion curves (Huhtanen and Hristov, 2001; Lund et al., 2006), or from indigestible NDF passage rate estimated by rumen evacuation technique (Huhtanen et al., 2006).

The assumptions of the NRC (2001) protein degradability model clearly overestimate differences between feeds in the supply of digestible RUP. The effect of the model assumptions on digestible RUP supply can be demonstrated using a 2-compartment model for fraction B (residence time 15 + 20 h) and assuming K_d and K_p values of 1.50 and 0.15/h for fraction A. Concentrations of fractions A, B, and C for 2 example forages were assumed to be 400, 550, and 50 and 600, 350, and 50 g/kg, respectively. Values of 0.08 and 0.05/h were used as K_d and K_p for fraction B. The difference in the flow of digestible RUP between the 2 example forages decreased from 79 to 19 g/kg when the escape of fraction A and selective retention of fraction B were included in the model. This example clearly indicates that the NRC (2001) model assumptions used to calculate the supply of RUP overestimate differences between feeds. The smaller effect of feed N fraction A on the supply of digestible RUP is in line with data from a recent meta-analysis of the effects of silage soluble N on MPY (Huhtanen et al., 2008a).

The relatively small MPY responses to reduced ruminal CP degradability observed in this meta-analysis may be related in part to a potentially lower intestinal digestibility and microbial protein flow with protein supplements treated for reduced ruminal degradability. Lebzien et al. (1996) analyzed 189 diets from studies conducted with fistulated dairy cows and concluded that duodenal flow of digestible protein is mostly dependent on intake of DOM or ME and nonurea CP. Inclusion of protein degradability in the model did not considerably improve prediction. The negative effect of increased undegradability of dietary CP on microbial protein outflow from the rumen was suggested by Voigt and Piatkowski (1987) and confirmed in a meta-analysis by Ipharraguerre and Clark (2005). The latter authors reported a 7% reduction in microbial protein outflow from the rumen with increasing RUP intake. Huhtanen (2005) reported that digestibility of incremental CP derived from heat-treated rapeseed expeller was lower (0.82 vs. 0.92) than that of solvent-extracted rapeseed meal. The analysis was based on the Lucas test of 24 and 38 diets containing either rapeseed expeller or extracted rapeseed meal.

The current meta-analysis clearly demonstrates the negative effect of increasing CP intake on MNE in dairy cows. Protein is usually overfed in US dairy herds (Jonker et al., 2002; Hristov et al., 2006). The common belief is that increasing dietary CP leads to increased MY. Examination of 31 publications (from the *Journal of Dairy Science*, volumes 78 to 91; 1995 to March 2008; list provided as a data supplement online: http://jds.fass.org/content/vol92/issue7/) in which CP level in the diet was a main effect showed increased MPY with increasing dietary CP in only 7 experiments (or 22%). In 5 of these 7 experiments, the effect on MPY was through increased DMI. In only 2 trials was MPY significantly increased with increasing dietary CP concentration, whereas there was no significant effect on DMI. In one of these publications (Metcalf et al., 1996), the control diet had extremely low CP (11.3%; there was no difference in MPY between the 15.4 and 20.1% CP diets). In the other study in which increased dietary CP resulted in greater MPY (Vagnoni and Broderick, 1997), diets apparently deficient in MP were supplemented with fish meal. Thus, it appears that there is no advantage of increasing dietary CP on the efficiency of conversion of dietary N in milk protein (Hristov et al., 2004). Although increasing dietary CP concentration can increase MPY, responses are usually relatively small, MNE is reduced (Huhtanen et al., 2008b), and urinary N excretion is increased.

The quadratic MP model performed best in cross-validation of the models in terms of smallest MSPE, mean, and slope biases. The linear MP model and TDN-CP models had considerable slope biases; the equation derived from the NA data underestimated MPY responses of NE cows and vice versa. The smaller linear regression coefficients of both MP (209 vs. 329 g/kg) and TDN models (55.9 vs. 75.5 g/kg) in the MPY models suggest than the NA cows were on a higher plane of nutrition, and therefore responded less to increases in nutrient supply. Intake of MPFeed was considerably greater in the NA data than in the NE data (0.99 vs. 0.60 kg/d), and the low efficiency of the MPFeed utilization may have contributed to the difference in MP response between the data sets.

The validations of CP and CP-Degr models indicated that the NE cows had a higher MNE than predicted by models developed from the NA data. This may be related to greater CP concentration in the NA diets, and smaller MPY responses to dietary CP. The TDN (discounted) concentration was slightly greater in the NE than in the NA data (672 vs. 652 g/kg of DM), which could also have contributed to the greater MNE in the NE data. Indeed, when TDN (intake or concentration) was included in the CP models, prediction errors decreased markedly and random error accounted for most of the error variance. Consistent with the MPY and MNE models, CP degradability did not improve predictions in the cross-validation. Predictions by the TDN-CP MNE models, whether based on concentrations or intakes of these nutrients, were accurate (small bias) and precise, indicating that MNE can be predicted reliably from these 2 parameters for dairy cows fed different types of diets.

CONCLUSIONS

This meta-analysis confirmed the importance of not overfeeding dietary protein as the most efficient tool in reducing N losses from dairy operations. Ruminal degradability of dietary protein as predicted by NRC (2001) does not appear to be a significant factor in predicting milk protein yield and milk N efficiency in dairy cows. Data also indicated that increasing MY would increase milk N efficiency, provided that dietary CP concentration is not increased, but the effect is considerably smaller than the effect of reducing CP intake.

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